

Types of Malware

Malware, short for malicious software, refers to any program or code designed to harm, exploit, or otherwise compromise computer systems without the owner's consent. There are several distinct categories of malware, each with different behaviors and objectives. Viruses attach themselves to legitimate files or programs and spread when the infected file is executed, often causing damage to data or system functionality. Worms, unlike viruses, can replicate and spread across networks independently without requiring a host file, sometimes causing widespread disruption by consuming network bandwidth. Trojans disguise themselves as legitimate software to trick users into installing them, after which they may create backdoors for attackers to access the infected system remotely. Ransomware encrypts a victim's files and demands payment for their release, and has become one of the most financially damaging forms of malware in recent years, targeting hospitals, businesses, and government agencies alike. Spyware secretly monitors user activity, often collecting sensitive information such as login credentials or browsing habits without the user's knowledge. Adware, while generally less harmful, bombards users with unwanted advertisements and can degrade system performance. Rootkits are particularly dangerous because they are designed to hide their presence and grant attackers persistent, privileged access to a system while evading detection by security software. Effective protection against malware requires a combination of up-to-date antivirus software, regular system patching to close known vulnerabilities, cautious online behavior, and network monitoring tools capable of detecting unusual activity that may indicate an infection.